# ANNEX I. Kyrgyz Republic: Debt Sustainability Analysis Update

*Based on the Debt Sustainability Analysis update, the Kyrgyz Republic remains at a moderate risk of debt distress. The debt outlook has improved slightly, but stress tests suggest that country is still vulnerable to large exogenous shocks. Fiscal consolidation and prudent use of concessional loans, in particular in the energy sector, are key for keeping the debt outlook sustainable and improving debt ratios in the medium term.*

1. **This joint Bank-Fund debt sustainability analysis (DSA) update by Fund and Bank staffs reconfirms the moderate risk rating found in the last DSA (country report No. 11/155).** The updated assessment incorporates: (i) the revised data on private sector debt since 2007; (ii) the higher borrowing for energy infrastructure (US$620 million compared to US$450 million expected previously) but on concessional terms as opposed to slightly nonconcessional terms in the last DSA; and (iii) a new loan from Turkey (US$100 million). Furthermore, the update factors in the completion of debt relief from Turkey (US$49.2 million) and Russia (US$188.9 million) in 2012 instead of 2011, and a write off of a US$300 million debt from Russia over ten years starting from 2016). The macroeconomic assumptions underlying the assessment have been revised to reflect temporarily lower growth in 2012, due to the deferral of a third of this year's gold production to 2013–14, and consequently higher growth in 2013–14. Despite the changes introduced in this update, with the long term assumptions remaining largely the same, there would not be any significant impact on the debt indicators.

**Table 1. Kyrgyz Republic: Underlying Long-Term DSA Assumptions**

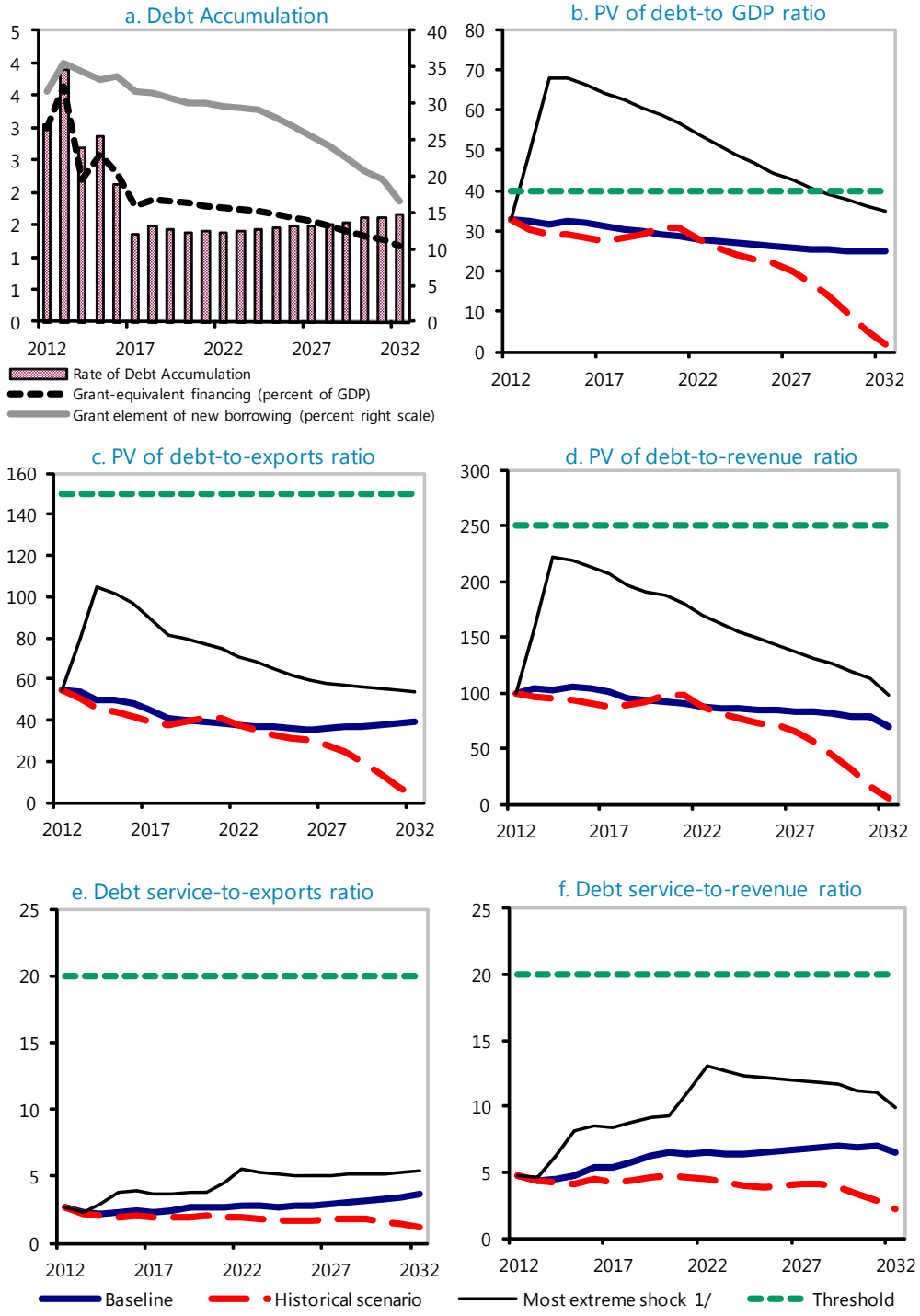
| | 2011 DSA Proj. | 2012 DSA Update Proj. |
|---|---|---|
| Real GDP growth (in percent) | 5.1 | 5.0 |
| Growth of exports of G&S (U.S. dollar terms, in percent) | 6.3 | 6.4 |
| Growth of imports of G&S (U.S. dollar terms, in percent) | 6.7 | 6.5 |
| Primary Deficit (percent of GDP) | 2.2 | 1.7 |
| Inflation rate | 4.1 | 4.0 |

Sources: Country authorities, and staff projections.

2. **The projected debt ratios under the baseline and alternative scenarios generally point to better outcomes than projected last year.** All external debt burden indicators in the baseline and alternative scenarios, except the ones related to the PV of debt to GDP in the most extreme shock (a combination of growth, export value, US dollar GDP deflator, and net non-debt creating flows as well as one time depreciation of som of 30 percent in 2013), are well below the thresholds.

Stress tests indicate that the Kyrgyz Republic still remains vulnerable to large external shocks and substantially less favorable assumptions. With regard to public debt, all debt indicators suggest a manageable debt outlook. The PV of debt to GDP ratio is gradually declining from 37 percent in 2012 to 32 percent in 2032. As in the previous DSA, alternative scenarios and stress test indicate that public debt is highly sensitive to shocks that reduce real GDP growth.

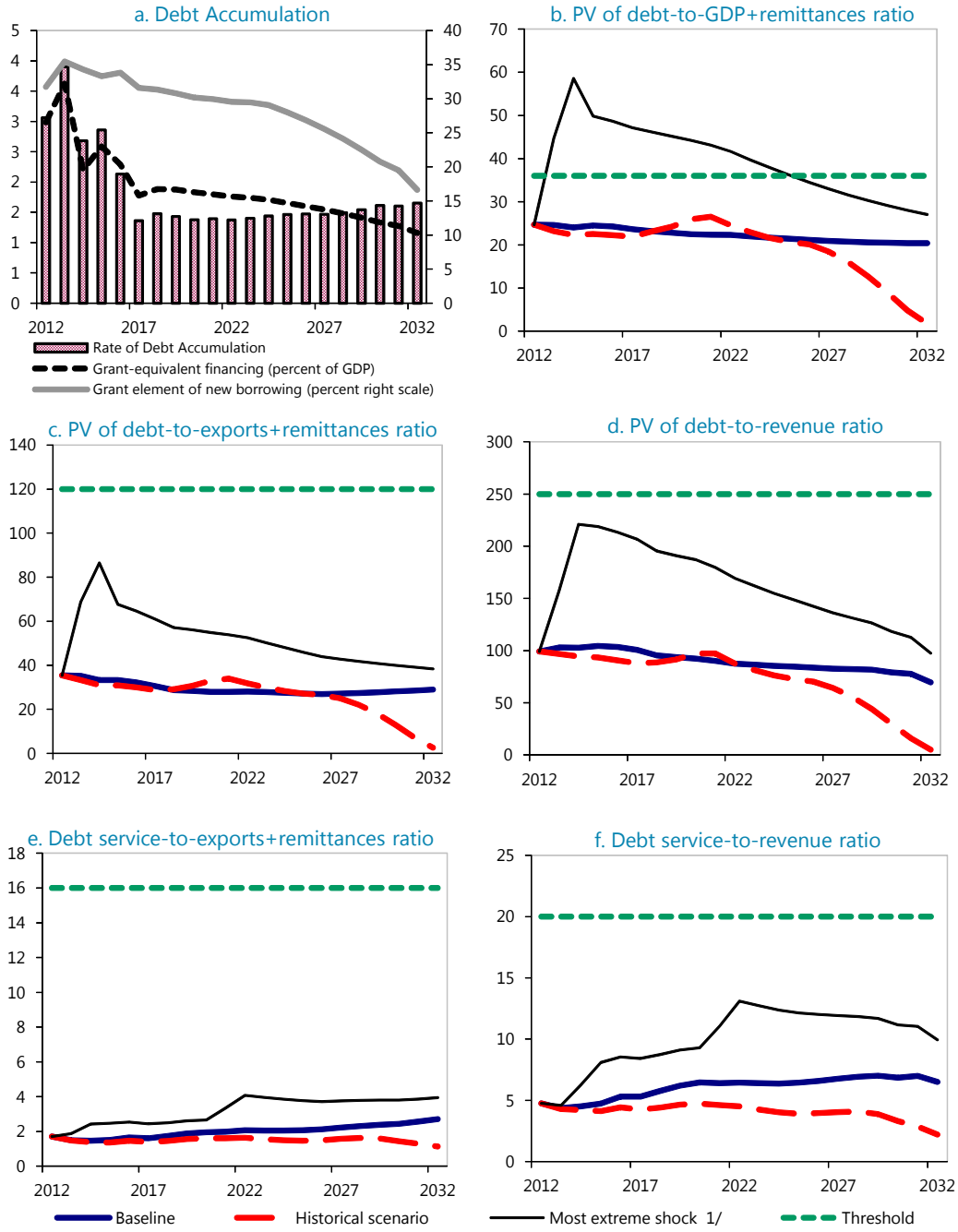
**Figure 1. Kyrgyz Republic: Indicators of Public and Publicly Guaranteed External Debt Under Alternatives Scenarios, 2012–32 1/**




Sources: Country authorities; and staff estimates and projections.

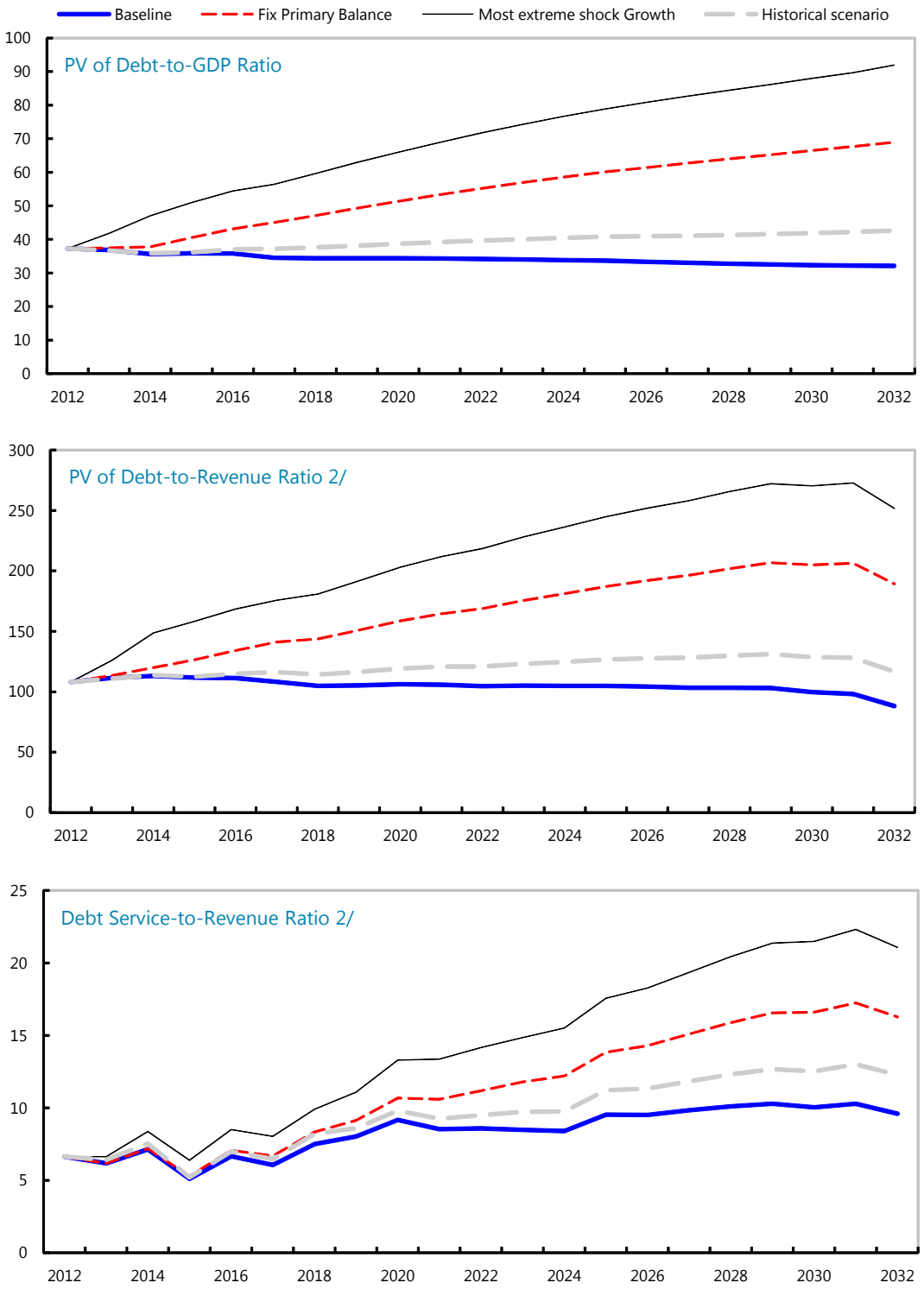
1/ The most extreme stress test is the test that yields the highest ratio in 2022. In figure b. it corresponds to a Combination shock; in c. to a Combination shock; in d. to a Combination shock; in e. to a Combination shock and in figure f. to a Combination shock

**Figure 2. Kyrgyz Republic: Indicators of Public and Publicly Guaranteed External Debt Under Alternatives Scenarios Including Remittances, 2012–32 1/**




Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2022. In figure b. it corresponds to a Non-debt flows shock; in c. to a Non-debt flows shock; in d. to a Combination shock; in e. to a Non-debt flows shock and in figure f. to a Combination shock

**Figure 3. Kyrgyz Republic: Indicators of Public Debt Under Alternative Scenarios, 2012–32 1/**




Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2022.

2/ Revenues are defined inclusive of grants.

## Table 1. Kyrgyz Republic: Public Sector Debt Sustainability Framework, Baseline Scenario, 2009–32

(In percent of GDP, unless otherwise indicated)

| | Actual | | | | | Estimate | Projections | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2009 | 2010 | 2011 | Average 6/ | Standard Deviation 6/ | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2012-17 Average | 2022 | 2032 | 2018-32 Average |
| **Public sector debt 1/** | 58.0 | 60.3 | 52.4 | | | 52.0 | 51.3 | 49.3 | 49.5 | 49.1 | 47.1 | | 44.8 | 40.7 | |
| o/w foreign-currency denominated | 55.1 | 57.4 | 49.2 | | | 48.1 | 47.4 | 45.7 | 46.6 | 45.9 | 44.1 | | 38.7 | 33.7 | |
| Change in public sector debt | 9.5 | 2.3 | -7.9 | | | -0.4 | -0.7 | -2.0 | 0.2 | -0.4 | -2.0 | | -0.4 | -0.3 | |
| Identified debt-creating flows | 5.7 | 4.5 | -7.7 | | | 4.0 | -1.2 | -1.9 | -0.7 | -1.4 | -2.1 | | -0.6 | -0.3 | |
| Primary deficit | 3.0 | 5.2 | 3.8 | 2.3 | 1.9 | 5.0 | 4.3 | 3.1 | 1.8 | 1.5 | 0.7 | 2.7 | 1.7 | 1.6 | 1.7 |
| Revenue and grants | 32.1 | 30.5 | 33.3 | | | 34.4 | 33.0 | 31.5 | 32.1 | 32.2 | 32.0 | | 32.7 | 36.4 | |
| *of which: grants* | 5.1 | 2.8 | 3.2 | | | 1.5 | 1.6 | 0.8 | 1.1 | 1.0 | 1.0 | | 0.8 | 0.6 | |
| Primary (noninterest) expenditure | 35.1 | 35.7 | 37.2 | | | 39.5 | 37.3 | 34.6 | 33.8 | 33.7 | 32.7 | | 34.4 | 38.0 | |
| Automatic debt dynamics | 2.6 | -0.6 | -11.4 | | | -1.0 | -5.1 | -4.5 | -2.2 | -2.7 | -2.8 | | -2.3 | -1.9 | |
| Contribution from interest rate/growth differential | -1.0 | 0.7 | -3.1 | | | -0.4 | -4.2 | -3.7 | -1.8 | -2.6 | -2.6 | | -2.3 | -1.9 | |
| *of which: contribution from average real interest rate* | 0.4 | 0.4 | 0.2 | | | 0.1 | -0.1 | -0.2 | -0.2 | -0.2 | -0.2 | | -0.2 | 0.1 | |
| *of which: contribution from real GDP growth* | -1.4 | 0.3 | -3.2 | | | -0.5 | -4.1 | -3.6 | -1.6 | -2.4 | -2.3 | | -2.1 | -2.0 | |
| Contribution from real exchange rate depreciation | 3.6 | -1.3 | -8.4 | | | -0.6 | -1.0 | -0.7 | -0.4 | -0.1 | -0.2 | | ... | ... | |
| Other identified debt-creating flows | 0.1 | -0.1 | -0.1 | | | -0.1 | -0.3 | -0.6 | -0.2 | -0.2 | 0.0 | | 0.0 | 0.0 | |
| Privatization receipts (negative) | 0.1 | -0.1 | -0.1 | | | -0.1 | -0.3 | -0.6 | -0.2 | -0.2 | 0.0 | | 0.0 | 0.0 | |
| Recognition of implicit or contingent liabilities | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Other (specify, e.g. bank recapitalization) | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Residual, including asset changes /2 | 3.8 | -2.1 | -0.2 | | | -4.4 | 0.5 | 0.0 | 0.9 | 1.0 | 0.0 | | 0.2 | 0.0 | |
| **Other Sustainability Indicators** | | | | | | | | | | | | | | | |
| **PV of public sector debt** | ... | ... | 34.8 | | | 37.3 | 36.8 | 35.6 | 35.8 | 35.8 | 34.5 | | 34.2 | 32.1 | |
| o/w foreign-currency denominated | ... | ... | 31.6 | | | 33.4 | 33.0 | 32.0 | 32.8 | 32.6 | 31.6 | | 28.1 | 25.0 | |
| o/w external | ... | ... | 30.8 | | | 32.7 | 32.4 | 31.5 | 32.4 | 32.2 | 31.2 | | 27.9 | 24.9 | |
| PV of contingent liabilities (not included in public sector debt) | ... | ... | ... | | | ... | ... | ... | ... | ... | ... | | ... | ... | |
| Gross financing need 3/ | 6.6 | 8.7 | 7.4 | | | 9.0 | 8.2 | 7.3 | 5.0 | 5.0 | 4.1 | | 6.8 | 5.6 | |
| PV of public sector debt-to-revenue and grants ratio (in percent) | ... | ... | 104.5 | | | 108.2 | 111.6 | 113.1 | 111.8 | 111.3 | 108.1 | | 104.6 | 88.1 | |
| PV of public sector debt-to-revenue ratio (in percent) | ... | ... | 115.5 | | | 113.0 | 117.2 | 116.1 | 115.7 | 115.0 | 111.5 | | 107.3 | 89.5 | |
| o/w external 4/ | ... | ... | 102.2 | | | 99.2 | 103.1 | 102.7 | 104.6 | 103.4 | 100.7 | | 87.5 | 69.5 | |
| Debt service-to-revenue and grants ratio (in percent) 5/ | 7.1 | 7.5 | 7.0 | | | 6.6 | 6.2 | 7.1 | 5.1 | 6.7 | 6.1 | | 8.6 | 9.6 | |
| Debt service-to-revenue ratio (in percent) 5/ | 8.5 | 8.2 | 7.8 | | | 6.9 | 6.5 | 7.3 | 5.3 | 6.9 | 6.2 | | 8.8 | 9.8 | |
| Primary deficit that stabilizes the debt-to-GDP ratio | -6.5 | 2.8 | 11.7 | | | 5.5 | 5.0 | 5.1 | 1.6 | 2.0 | 2.8 | | 2.1 | 1.9 | |
| **Key macroeconomic and fiscal assumptions** | | | | | | | | | | | | | | | |
| Real GDP growth (in percent) | 2.9 | -0.5 | 5.7 | 4.1 | 3.6 | 1.0 | 8.5 | 7.5 | 3.3 | 5.0 | 5.0 | 5.1 | 5.0 | 5.0 | 5.0 |
| Average nominal interest rate on forex debt (in percent) | 1.2 | 1.1 | 1.2 | 1.3 | 0.3 | 1.2 | 1.2 | 1.1 | 1.2 | 1.2 | 1.2 | 1.2 | 1.5 | 2.1 | 1.6 |
| Average real interest rate on domestic debt (in percent) | 10.0 | 2.5 | -3.2 | -0.3 | 5.2 | 7.8 | 7.5 | 4.9 | 3.6 | 3.7 | 2.8 | 5.0 | 0.2 | 0.1 | 0.2 |
| Real exchange rate depreciation (in percent, + indicates depreciation) | 8.1 | -2.3 | -15.3 | -6.8 | 7.4 | -1.2 | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| Inflation rate (GDP deflator, in percent) | 4.0 | 10.0 | 17.3 | 9.7 | 6.8 | 5.8 | 7.4 | 6.8 | 6.0 | 5.4 | 5.5 | 6.1 | 4.0 | 4.0 | 4.0 |
| Growth of real primary spending (deflated by GDP deflator, in percent) | 0.3 | 0.0 | 0.1 | 0.1 | 0.1 | 0.1 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.1 | 0.2 | 0.1 |
| Grant element of new external borrowing (in percent) | ... | ... | ... | ... | ... | 31.8 | 35.5 | 34.3 | 33.3 | 33.8 | 31.6 | 33.4 | 29.5 | 16.6 | ... |

Sources: Country authorities; and staff estimates and projections.
1/ Covers general government and gross debt is used.
2/ Includes debt relief.
3/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
4/ Revenues excluding grants.
5/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

## Table 2. Kyrgyz Republic: Sensitivity Analysis for Key Indicators of Public Debt, 2012–32

| | Projections | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2022 | 2032 |
| **PV of Debt-to-GDP Ratio** | | | | | | | | |
| **Baseline** | 37 | 37 | 36 | 36 | 36 | 35 | 34 | 32 |
| **A. Alternative scenarios** | | | | | | | | |
| A1. Real GDP growth and primary balance are at historical averages | 37 | 37 | 36 | 36 | 37 | 37 | 40 | 43 |
| A2. Primary balance is unchanged from 2012 | 37 | 37 | 38 | 41 | 43 | 45 | 55 | 69 |
| A3. Permanently lower GDP growth 1/ | 37 | 37 | 37 | 38 | 39 | 39 | 47 | 72 |
| **B. Bound tests** | | | | | | | | |
| B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014 | 37 | 42 | 47 | 51 | 54 | 56 | 72 | 92 |
| B2. Primary balance is at historical average minus one standard deviations in 2013-2014 | 37 | 37 | 36 | 37 | 37 | 35 | 35 | 32 |
| B3. Combination of B1-B2 using one half standard deviation shocks | 37 | 38 | 39 | 42 | 44 | 46 | 58 | 74 |
| B4. One-time 30 percent real depreciation in 2013 | 37 | 50 | 47 | 46 | 46 | 44 | 42 | 38 |
| B5. 10 percent of GDP increase in other debt-creating flows in 2013 | 37 | 45 | 43 | 43 | 43 | 41 | 40 | 35 |
| **PV of Debt-to-Revenue Ratio 2/** | | | | | | | | |
| **Baseline** | 108 | 112 | 113 | 112 | 111 | 108 | 105 | 88 |
| **A. Alternative scenarios** | | | | | | | | |
| A1. Real GDP growth and primary balance are at historical averages | 108 | 111 | 114 | 113 | 115 | 116 | 121 | 117 |
| A2. Primary balance is unchanged from 2012 | 108 | 113 | 120 | 126 | 134 | 141 | 169 | 189 |
| A3. Permanently lower GDP growth 1/ | 108 | 113 | 116 | 118 | 120 | 120 | 142 | 198 |
| **B. Bound tests** | | | | | | | | |
| B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014 | 108 | 126 | 149 | 158 | 169 | 176 | 219 | 252 |
| B2. Primary balance is at historical average minus one standard deviations in 2013-2014 | 108 | 111 | 116 | 114 | 114 | 110 | 106 | 89 |
| B3. Combination of B1-B2 using one half standard deviation shocks | 108 | 115 | 122 | 129 | 138 | 143 | 177 | 203 |
| B4. One-time 30 percent real depreciation in 2013 | 108 | 151 | 149 | 145 | 142 | 138 | 127 | 105 |
| B5. 10 percent of GDP increase in other debt-creating flows in 2013 | 108 | 136 | 137 | 134 | 133 | 129 | 121 | 97 |
| **Debt Service-to-Revenue Ratio 2/** | | | | | | | | |
| **Baseline** | 7 | 6 | 7 | 5 | 7 | 6 | 9 | 10 |
| **A. Alternative scenarios** | | | | | | | | |
| A1. Real GDP growth and primary balance are at historical averages | 7 | 6 | 8 | 5 | 7 | 6 | 10 | 12 |
| A2. Primary balance is unchanged from 2012 | 7 | 6 | 7 | 5 | 7 | 7 | 11 | 16 |
| A3. Permanently lower GDP growth 1/ | 7 | 6 | 7 | 5 | 7 | 6 | 10 | 16 |
| **B. Bound tests** | | | | | | | | |
| B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014 | 7 | 7 | 8 | 6 | 8 | 8 | 14 | 21 |
| B2. Primary balance is at historical average minus one standard deviations in 2013-2014 | 7 | 6 | 7 | 5 | 7 | 6 | 9 | 10 |
| B3. Combination of B1-B2 using one half standard deviation shocks | 7 | 7 | 8 | 6 | 8 | 7 | 12 | 17 |
| B4. One-time 30 percent real depreciation in 2013 | 7 | 7 | 9 | 7 | 9 | 9 | 12 | 14 |
| B5. 10 percent of GDP increase in other debt-creating flows in 2013 | 7 | 6 | 8 | 6 | 7 | 7 | 10 | 11 |

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

## Table 3a. Kyrgyz Republic: External Debt Sustainability Framework, Baseline Scenario, 2009–32 1/

(In percent of GDP, unless otherwise indicated)

| | Actual | | | Historical Average 6/ | Standard Deviation 6/ | Projections | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2009 | 2010 | 2011 | | | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2012-2017 Average | 2022 | 2032 | 2018-2032 Average |
| **External debt (nominal) 1/** | **89.3** | **92.8** | **81.2** | | | **80.0** | **75.5** | **71.3** | **71.1** | **69.6** | **68.0** | | **63.8** | **73.5** | |
| o/w public and publicly guaranteed (PPG) | 54.1 | 56.5 | 48.4 | | | 47.5 | 46.8 | 45.2 | 46.1 | 45.5 | 43.7 | | 38.5 | 33.6 | |
| Change in external debt | 16.0 | 3.5 | -11.6 | | | -1.2 | -4.5 | -4.1 | -0.2 | -1.5 | -1.6 | | -1.3 | 1.2 | |
| Identified net debt-creating flows | 5.5 | -4.8 | -23.1 | | | 1.3 | -6.0 | -6.6 | -5.0 | -6.0 | -6.2 | | -3.3 | 2.7 | |
| **Non-interest current account deficit** | **1.6** | **5.6** | **5.8** | **2.4** | **6.2** | **7.1** | **4.6** | **2.7** | **0.8** | **0.7** | **0.2** | | **1.3** | **5.3** | 1.4 |
| Deficit in balance of goods and services | 24.4 | 29.0 | 28.3 | | | 35.9 | 31.0 | 27.1 | 27.7 | 27.7 | 27.0 | | 25.1 | 26.3 | |
| Exports | 54.5 | 52.7 | 58.8 | | | 59.8 | 60.4 | 63.6 | 65.2 | 67.4 | 70.1 | | 74.1 | 63.8 | |
| Imports | 78.9 | 81.7 | 87.2 | | | 95.7 | 91.5 | 90.7 | 92.9 | 95.1 | 97.1 | | 99.2 | 90.2 | |
| Net current transfers (negative = inflow) | -25.8 | -29.0 | -31.0 | -21.6 | 8.9 | -32.4 | -31.5 | -31.1 | -32.2 | -32.4 | -31.9 | | -25.4 | -22.4 | -25.3 |
| o/w official | -4.2 | -1.6 | -1.4 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | -0.4 | -0.2 | |
| Other current account flows (negative = net inflow) | 2.9 | 5.6 | 8.4 | | | 3.6 | 5.1 | 6.7 | 5.3 | 5.4 | 5.1 | | 1.6 | 1.3 | |
| **Net FDI (negative = inflow)** | **-4.0** | **-9.1** | **-11.7** | **-5.5** | **3.5** | **-7.0** | **-6.4** | **-6.1** | **-5.5** | **-5.3** | **-5.2** | | **-4.3** | **-4.4** | -4.4 |
| **Endogenous debt dynamics 2/** | **8.0** | **-1.3** | **-17.1** | | | **1.1** | **-4.2** | **-3.2** | **-0.3** | **-1.3** | **-1.2** | | **-0.3** | **1.8** | |
| Contribution from nominal interest rate | 1.0 | 0.8 | 0.5 | | | 1.9 | 1.8 | 1.8 | 1.9 | 2.0 | 2.1 | | 2.8 | 5.2 | |
| Contribution from real GDP growth | -2.3 | 0.4 | -4.3 | | | -0.8 | -6.0 | -5.1 | -2.2 | -3.3 | -3.2 | | -3.0 | -3.4 | |
| Contribution from price and exchange rate changes | 9.3 | -2.5 | -13.4 | | | ... | ... | ... | ... | ... | ... | | ... | ... | |
| **Residual (3-4) 3/** | **10.5** | **8.3** | **11.5** | | | **-2.4** | **1.5** | **2.5** | **4.7** | **4.5** | **4.5** | | **2.0** | **-1.4** | |
| o/w exceptional financing | 0.0 | 0.0 | 0.0 | | | -0.8 | -2.7 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| PV of external debt 4/ | ... | ... | 63.6 | | | 65.2 | 61.0 | 57.7 | 57.4 | 56.3 | 55.4 | | 53.1 | 64.9 | |
| In percent of exports | ... | ... | 108.1 | | | 109.1 | 101.0 | 90.6 | 88.1 | 83.5 | 79.1 | | 71.7 | 101.7 | |
| **PV of PPG external debt** | **...** | **...** | **30.8** | | | **32.7** | **32.4** | **31.5** | **32.4** | **32.2** | **31.2** | | **27.9** | **24.9** | |
| **In percent of exports** | **...** | **...** | **52.4** | | | **54.7** | **53.6** | **49.5** | **49.7** | **47.7** | **44.5** | | **37.6** | **39.0** | |
| **In percent of government revenues** | **...** | **...** | **102.2** | | | **99.2** | **103.1** | **102.7** | **104.6** | **103.4** | **100.7** | | **87.5** | **69.5** | |
| **Debt service-to-exports ratio (in percent)** | **28.2** | **19.1** | **8.9** | | | **13.0** | **14.5** | **12.7** | **13.0** | **13.1** | **11.8** | | **17.4** | **36.0** | |
| **PPG debt service-to-exports ratio (in percent)** | **3.3** | **3.6** | **2.8** | | | **2.6** | **2.3** | **2.2** | **2.3** | **2.4** | **2.3** | | **2.8** | **3.7** | |
| **PPG debt service-to-revenue ratio (in percent)** | **6.7** | **6.9** | **5.5** | | | **4.8** | **4.4** | **4.5** | **4.7** | **5.3** | **5.3** | | **6.4** | **6.5** | |
| Total gross financing need (Billions of U.S. dollars) | 0.6 | 0.3 | 0.0 | | | 0.5 | 0.5 | 0.4 | 0.3 | 0.4 | 0.3 | | 1.3 | 6.4 | |
| Non-interest current account deficit that stabilizes debt ratio | -14.4 | 2.2 | 17.4 | | | 8.3 | 9.2 | 6.8 | 1.0 | 2.2 | 1.8 | | 2.6 | 4.0 | |
| **Key macroeconomic assumptions** | | | | | | | | | | | | | | | |
| Real GDP growth (in percent) | 2.9 | -0.5 | 5.7 | 4.1 | 3.6 | 1.0 | 8.5 | 7.5 | 3.3 | 5.0 | 5.0 | 5.1 | 5.0 | 5.0 | 5.0 |
| GDP deflator in US dollar terms (change in percent) | -11.3 | 2.8 | 16.9 | 10.5 | 10.5 | 3.5 | 4.2 | 3.7 | 2.9 | 2.3 | 2.4 | 3.2 | 2.0 | 2.0 | 2.0 |
| Effective interest rate (percent) 5/ | 1.2 | 0.9 | 0.7 | 1.3 | 0.4 | 2.5 | 2.6 | 2.7 | 2.8 | 3.0 | 3.2 | 2.8 | 4.5 | 7.7 | 5.5 |
| Growth of exports of G&S (US dollar terms, in percent) | -7.8 | -1.1 | 37.9 | 21.3 | 18.0 | 6.3 | 14.3 | 17.3 | 8.9 | 11.0 | 11.9 | 11.6 | 6.4 | 4.8 | 6.4 |
| Growth of imports of G&S (US dollar terms, in percent) | -22.3 | 6.0 | 31.7 | 26.4 | 23.2 | 14.7 | 8.1 | 10.6 | 8.8 | 9.9 | 9.9 | 10.3 | 7.3 | 3.7 | 6.5 |
| Grant element of new public sector borrowing (in percent) | ... | ... | ... | ... | ... | 31.8 | 35.5 | 34.3 | 33.3 | 33.8 | 31.6 | 33.4 | 29.5 | 16.6 | 26.3 |
| Government revenues (excluding grants, in percent of GDP) | 27.1 | 27.7 | 30.2 | | | 33.0 | 31.4 | 30.7 | 31.0 | 31.1 | 31.0 | | 31.9 | 35.8 | 31.8 |
| Aid flows (in Billions of US dollars) 7/ | 0.6 | 0.3 | 0.3 | | | 0.2 | 0.3 | 0.2 | 0.3 | 0.2 | 0.2 | | 0.3 | 0.4 | |
| o/w Grants | 0.2 | 0.1 | 0.2 | | | 0.1 | 0.1 | 0.1 | 0.1 | 0.1 | 0.1 | | 0.1 | 0.2 | |
| o/w Concessional loans | 0.4 | 0.1 | 0.1 | | | 0.1 | 0.2 | 0.2 | 0.2 | 0.2 | 0.1 | | 0.2 | 0.3 | |
| Grant-equivalent financing (in percent of GDP) 8/ | ... | ... | ... | | | 3.0 | 3.6 | 2.2 | 2.6 | 2.3 | 1.8 | | 1.8 | 1.2 | 1.6 |
| Grant-equivalent financing (in percent of external financing) 8/ | ... | ... | ... | | | 47.6 | 49.6 | 45.4 | 46.4 | 48.4 | 50.7 | | 43.6 | 28.2 | 39.6 |
| *Memorandum items:* | | | | | | | | | | | | | | | |
| Nominal GDP (Billions of US dollars) | 4.7 | 4.8 | 5.9 | | | 6.2 | 7.0 | 7.8 | 8.3 | 8.9 | 9.6 | | 13.5 | 26.6 | |
| Nominal dollar GDP growth | -8.7 | 2.4 | 23.5 | | | 4.5 | 13.1 | 11.4 | 6.3 | 7.4 | 7.6 | 8.4 | 7.1 | 7.1 | 7.1 |
| PV of PPG external debt (in Billions of US dollars) | | | 1.8 | | | 2.0 | 2.2 | 2.4 | 2.6 | 2.8 | 2.9 | | 3.7 | 6.6 | |
| (PVt-PVt-1)/GDPt-1 (in percent) | | | | | | 3.1 | 3.9 | 2.7 | 2.9 | 2.1 | 1.4 | 2.7 | 1.4 | 1.7 | 1.5 |
| Gross workers' remittances (Billions of US dollars) | 1.0 | 1.3 | 1.8 | | | 2.0 | 2.2 | 2.4 | 2.7 | 2.9 | 3.1 | | 3.4 | 5.9 | |
| PV of PPG external debt (in percent of GDP + remittances) | ... | ... | 23.8 | | | 24.7 | 24.6 | 24.0 | 24.5 | 24.3 | 23.6 | | 22.3 | 20.4 | |
| PV of PPG external debt (in percent of exports + remittances) | ... | ... | 34.9 | | | 35.4 | 35.2 | 33.3 | 33.2 | 32.2 | 30.6 | | 28.1 | 28.9 | |
| Debt service of PPG external debt (in percent of exports + remittances) | ... | ... | 1.9 | | | 1.7 | 1.5 | 1.5 | 1.5 | 1.7 | 1.6 | | 2.1 | 2.7 | |

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes and accounts for the financing gap, which creates debt but is not included in the current or financial account.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

## Table 3b. Kyrgyz Republic: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2012–32

(In percent)

| | Projections | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | **2022** | 2032 |
| **PV of debt-to GDP ratio** | | | | | | | | |
| **Baseline** | 33 | 32 | 32 | 32 | 32 | 31 | **28** | 25 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2012-2032 1/ | 33 | 30 | 29 | 29 | 28 | 27 | **28** | 2 |
| A2. New public sector loans on less favorable terms in 2012-2032 2 | 33 | 34 | 34 | 36 | 36 | 36 | **36** | 41 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2013-2014 | 33 | 34 | 36 | 37 | 37 | 35 | **32** | 28 |
| B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/ | 33 | 37 | 46 | 46 | 45 | 44 | **38** | 28 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014 | 33 | 33 | 34 | 34 | 34 | 33 | **30** | 26 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/ | 33 | 50 | 66 | 66 | 64 | 62 | **52** | 33 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 33 | 49 | 68 | 68 | 66 | 64 | **54** | 35 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/ | 33 | 45 | 44 | 45 | 44 | 43 | **39** | 34 |
| **PV of debt-to-exports ratio** | | | | | | | | |
| **Baseline** | 55 | 54 | 50 | 50 | 48 | 44 | **38** | 39 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2012-2032 1/ | 55 | 50 | 45 | 44 | 42 | 39 | **38** | 3 |
| A2. New public sector loans on less favorable terms in 2012-2032 2 | 55 | 56 | 53 | 55 | 54 | 51 | **49** | 65 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2013-2014 | 55 | 53 | 49 | 49 | 47 | 44 | **37** | 38 |
| B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/ | 55 | 67 | 90 | 89 | 85 | 79 | **64** | 55 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014 | 55 | 53 | 49 | 49 | 47 | 44 | **37** | 38 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/ | 55 | 83 | 104 | 101 | 96 | 89 | **70** | 52 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 55 | 79 | 104 | 102 | 96 | 89 | **71** | 54 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/ | 55 | 53 | 49 | 49 | 47 | 44 | **37** | 38 |
| **PV of debt-to-revenue ratio** | | | | | | | | |
| **Baseline** | 99 | 103 | 103 | 105 | 103 | 101 | **88** | 69 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2012-2032 1/ | 99 | 97 | 94 | 93 | 91 | 88 | **87** | 5 |
| A2. New public sector loans on less favorable terms in 2012-2032 2 | 99 | 107 | 110 | 115 | 116 | 115 | **114** | 116 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2013-2014 | 99 | 110 | 117 | 119 | 118 | 114 | **100** | 79 |
| B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/ | 99 | 116 | 149 | 149 | 146 | 142 | **119** | 78 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014 | 99 | 106 | 109 | 111 | 110 | 107 | **93** | 74 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/ | 99 | 160 | 215 | 213 | 207 | 201 | **163** | 92 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 99 | 157 | 221 | 219 | 213 | 207 | **169** | 98 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/ | 99 | 143 | 142 | 144 | 143 | 139 | **121** | 96 |

## Table 3b. Kyrgyz Republic: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2012–32 (continued)

(In percent)

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **Debt service-to-exports ratio** | | | | | | | | |
| **Baseline** | 3 | 2 | 2 | 2 | 2 | 2 | **3** | 4 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2012-2032 1/ | 3 | 2 | 2 | 2 | 2 | 2 | **2** | 1 |
| A2. New public sector loans on less favorable terms in 2012-2032 2 | 3 | 2 | 2 | 2 | 3 | 3 | **3** | 4 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2013-2014 | 3 | 2 | 2 | 2 | 2 | 2 | **3** | 4 |
| B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/ | 3 | 3 | 3 | 4 | 4 | 4 | **5** | 5 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014 | 3 | 2 | 2 | 2 | 2 | 2 | **3** | 4 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/ | 3 | 2 | 3 | 4 | 4 | 4 | **5** | 5 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 3 | 2 | 3 | 4 | 4 | 4 | **6** | 5 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/ | 3 | 2 | 2 | 2 | 2 | 2 | **3** | 4 |
| **Debt service-to-revenue ratio** | | | | | | | | |
| **Baseline** | 5 | 4 | 5 | 5 | 5 | 5 | **6** | 7 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2012-2032 1/ | 5 | 4 | 4 | 4 | 4 | 4 | **5** | 2 |
| A2. New public sector loans on less favorable terms in 2012-2032 2 | 5 | 4 | 5 | 5 | 6 | 6 | **7** | 8 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2013-2014 | 5 | 5 | 5 | 5 | 6 | 6 | **7** | 7 |
| B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/ | 5 | 4 | 5 | 6 | 6 | 6 | **9** | 8 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014 | 5 | 5 | 5 | 5 | 6 | 6 | **7** | 7 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/ | 5 | 4 | 6 | 8 | 8 | 8 | **13** | 9 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 5 | 5 | 6 | 8 | 9 | 8 | **13** | 10 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/ | 5 | 6 | 6 | 7 | 7 | 7 | **9** | 9 |
| *Memorandum item:* | | | | | | | | |
| Grant element assumed on residual financing (i.e., financing required above baseline) 6/ | 21 | 21 | 21 | 21 | 21 | 21 | **21** | 21 |

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.